Chapter 4: Continuous Random Variables and Probability Distributions

PART A:

- CONTINUOUS RANDOM VARIABLE AND DISTRIBUTION
- EXPECTED VALUE, VARIANCE, STANDARD DEVIATION

Learning Objectives:

At the end of this chapter, the students would be able to:

- ✓ Understand the concept of a continuous random variable and probability distribution
- ✓ Be able to compute the expected value, variance, and standard deviation for a continuous random variable
- ✓ Be able to compute probabilities using a Uniform distribution, Normal distribution, Normal approximation to the Binomial distribution and Normal approximation to the Poisson distribution
- ✓ Continuous Probability Distributions (Probability Mass Functions, Mathematical Expectations)
- ✓ Special Continuous Probability Distributions (Uniform Normal; Normal Approximation to the Binomial Distribution; Normal Approximation to the Poisson Distribution)

4.1 Continuous Random Variable

Remember that random variables may be classified as:

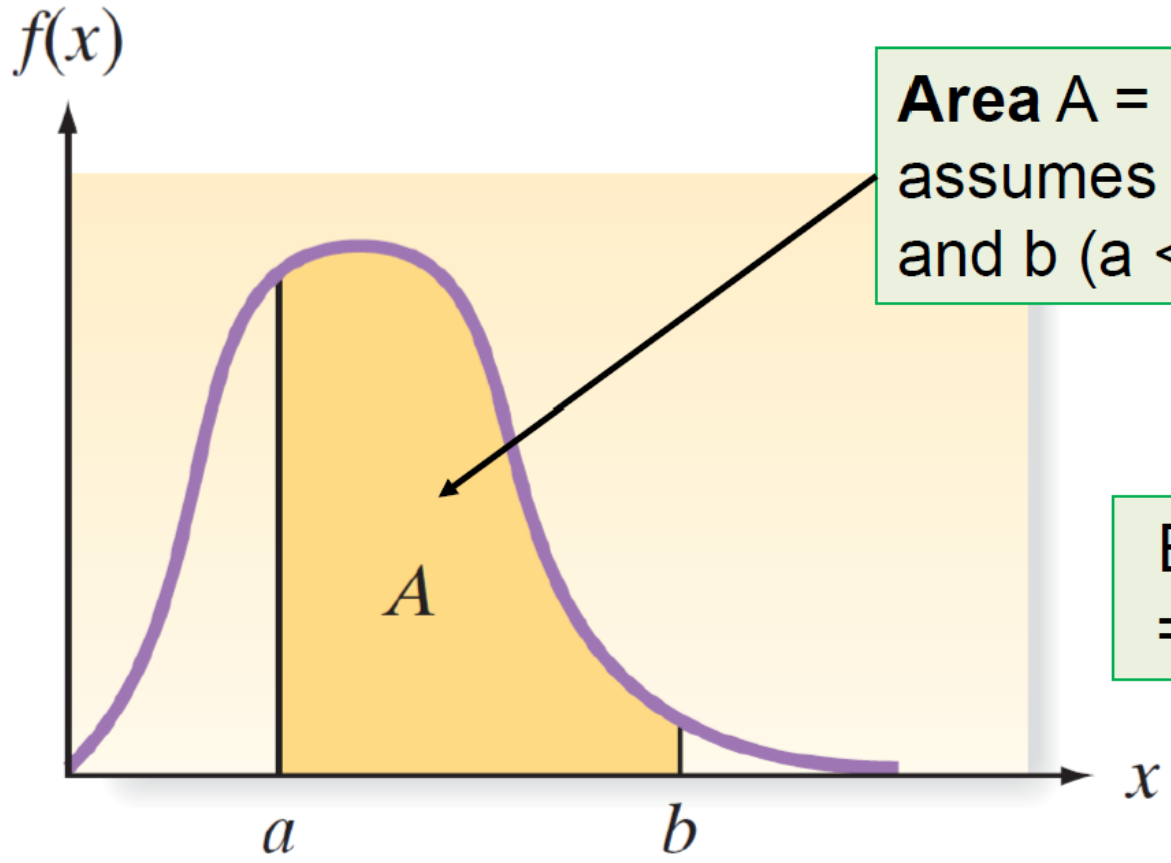
✓ **Discrete** The random variable assumes a countable number of distinct values.

✓ **Continuous** The random variable is characterized by (infinitely) uncountable values within any interval.

4.1 Continuous Random Variable

A continuous random variable X is defined by its **probability density function, f(x),** with for all value of X which are valid in its interval. If X is a continuous random variable with probability density function, f(x) valid in the **interval [a, b]** then that is the **area under the curve y = f(x) between x = a and x = b is 1**.

4.1 Continuous Random Variable



Area A = **Probability** that X assumes a value between a and b (a < X < b)

Entire area under f(x) = Total probability = 1

4.1 Continuous Random Variable

- When computing probabilities for a continuous random variable, keep in mind that $P(X = x) = 0$.Continuous random variables can take an **infinite number of possible values** within any range, no matter how small the range.
- The probability is spread out over all possible values, and the probability of hitting a specific, single point is essentially **negligible**.
- Instead of focusing on the probability of exact values, we calculate the probability over **intervals** (e.g. between 6 and 8 seconds).

4.1 Continuous Random Variable

- Example:

It is impossible to figure out the probability that the download time for a video is exactly 7 seconds. Why not?

- It is unlikely that the download time is *exactly 7.0000..* seconds.
- It could be 6.9999 seconds or 7.0001 seconds.
- Therefore, we can only determine the probability that the download time is between 6 to 8 seconds.

4.1 Continuous Random Variable

- Types of continuous probability distribution:

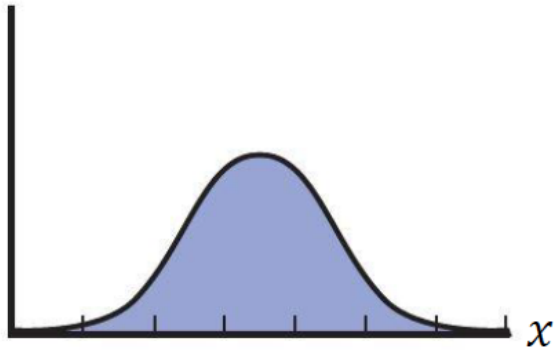


Figure 1:
Normal distribution

- Mean = Median
- $-\infty < X < \infty$

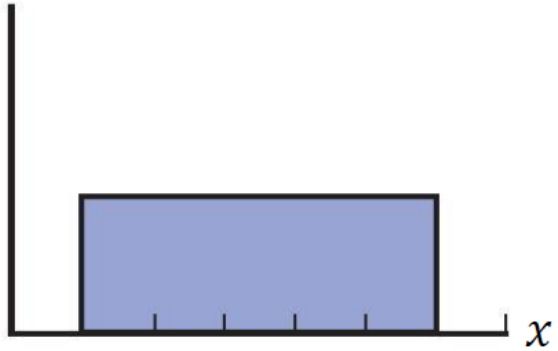


Figure 2:
Uniform distribution

- Mean = Median
- $a < X < b$

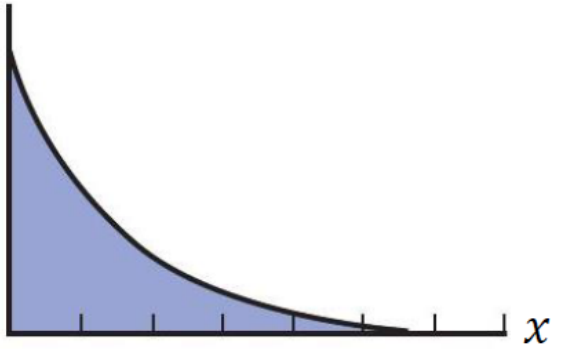


Figure 3:
Exponential distribution

- Mean > Median
- $0 < X < \infty$

4.1 Continuous Random Variable

The probability density function must satisfy several conditions:

i. $f(x) \geq 0$

ii. $\int_{-\infty}^{\infty} f(x)dx = 1$

iii. $P(a \leq x \leq b) = P(a \leq x \leq b) = P(a \leq x \leq b) = P(a \leq x \leq b) = \int_{a}^{b} f(x)dx$

4.1 Continuous Random Variable

For power rule:

$$\int x^n dx = \frac{x^{n+1}}{n+1} + c, n \neq -1$$

$$\int x^{-1} dx = \ln|x| + c, n \neq -1$$

4.1 Continuous Random Variable

For exponential:

$$\int e^x dx = e^x + c$$

$$\int a^x dx = \frac{a^x}{\ln a} + c$$

4.1 Continuous Random Variable

Example 1 (from textbook Introduction to Probability and Statistics, UUM, (2021); pg. 84)

A continuous random variable Y has the probability density function

$$f(y) = \begin{cases} \frac{y^2}{3}, & -1 \leq y \leq 2 \\ 0 & elsewhere \end{cases}$$

Calculate:

(i) Verify that $\int_{-\infty}^{\infty} f(y)dy = 1$

(ii) Find $P(0 < y < 1)$

(iii) Find $P(0 < y \leq 2)$

4.1 Continuous Random Variable

Example 2 (from textbook Introduction to Probability and Statistics, UUM, (2021); pg. 85)

A continuous random variable X has the probability density function

$$f(x) = \begin{cases} x & 0 < x < 1 \\ 2 - x & 1 \leq x < 2 \\ 0 & elsewhere \end{cases}$$

Find $P(X > 0.2)$.

Expected value, Variance, and Standard deviation

Summary measures for a random variable include the

- ✓ Mean (Expected Value)
- ✓ Variance
- ✓ Standard Deviation

Expected value / Mean

The expected value (mean) or average of X, being the random variable with probability distribution f(x) is:

$$\mu = E(X) = \int_{-\infty}^{\infty} xf(x)dx \text{ , if X is continuous}$$

**or when f(x)=X^2 so $E(X^2) = \int_{-\infty}^{\infty} x^2 f(x)dx$

Expected value / Mean

For any constant a and b:

i. $E(a) = a$

ii. $E(aX) = aE(X)$

iii. $E(X \pm b) = E(X) \pm E(b)$

Variance, and Standard deviation

- Formula for Variance (Var) and Standard deviation (SD):

$$\sigma^2 = Var\,(X) = E\left(X^2\right) - E(X)^2$$

$$\sigma^2 = Var\,(X) = E\left(X^2\right) - \mu^2$$

$E(X)$; the mean

$$SD(X) = \sqrt{Var(X)}$$

$$SD(X) = \sigma = \sqrt{\sigma^2}$$

Variance, and Standard deviation

- For any constant a and b:

$$Var\,(a) = 0$$
$$Var\,(aX) = a^2 Var(X)$$
$$Var\,(aX \pm b) = a^2 Var(X)$$

Mean, Variance and standard deviation

Example 3 (from textbook Introduction to Probability and Statistics, UUM, (2021); pg. 87)

Given that the probability density function for X.

$$f(x) = \begin{cases} \frac{3}{16}(4 - x^2), & 0 \leq x \leq 2 \\ 0 & elsewhere \end{cases}$$

Calculate:

(i) $E(X)$

(ii) $Var\ (X)$

(iii) SD (X)

(iv) $E\left(X^2 + 5X - 3\right)$

(v) Var $(3X - 7)$

EXERCISES

Exercise 4.2 from textbook Introduction to Probability and Statistics, UUM, (2021); pg. 88 – 89

Given that the probability density function for X.

$$f(x) = \begin{cases} 2(1-x), & 0 < x < 1 \\ 0 & elsewhere \end{cases}$$

Calculate:

(i) $E(X)$

(ii) $E(X^2)$

(iii) $E(2X+5)$

(iv) Var $(3+2X)$